

# Zeel's Brain v1 - DSA Revision Tracker

**Zeel's Brain v1** is a high-performance web application designed to help developers master Data Structures and Algorithms (DSA) through **Spaced Repetition**. It acts as a digital second brain, ensuring you never forget a problem once you've solved it.

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## The Problem: The Forgetting Curve

Most developers solve a complex DSA problem today but struggle to recall the logic 2 weeks later. Traditional note-taking isn't enough. **Zeel's Brain v1** solves this by scheduling revisions at scientifically optimized intervals.

## Key Features

### Spaced Repetition System (SRS)

The core of the app. When you solve a problem, the system automatically schedules your next revision.

- **Intervals:** 2, 7, 15, 30, 60, 120, and 240 days.
- **Auto-Progression:** Every time you revise, the problem moves to the next interval.
- **Completion:** After 7 successful revisions, the problem is marked as 'Mastered'.

### Interactive Dashboard

A premium UI built for productivity:

- **Quick Stats:** Track total problems solved, due revisions, and your current streak.
- **Platform Breakdown:** Visual insights into where you solve problems (LeetCode, Codeforces, GFG, etc.).
- **Revision Radar:** Easily see what's due today and what's coming up.

### Premium User Experience

- **Glassmorphism Design:** A sleek, modern aesthetic with translucent layers.
- **Micro-animations:** Smooth transitions powered by GSAP and Framer Motion for a fluid feel.
- **Dark Mode Optimized:** Designed to be easy on the eyes during late-night coding sessions.

### Persistence

- **Local First:** All your data is saved securely in your browser's `LocalStorage`. No database setup required—just open and code.
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## Tech Stack

- **Frontend Framework:** [React 19](#)
  - **Build Tool:** [Vite](#)
  - **Styling:** Vanilla CSS (Custom tokens and Glassmorphism)
  - **Animation:** [GSAP](#) & [Framer Motion](#)
  - **Icons:** [Lucide React](#)
  - **State Management:** React Context API
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## Project Architecture

The project follows a clean, modular structure:

- `src/context/RevisionContext.jsx` : The central "brain" of the app. Handles the logic for adding questions, marking revisions, and calculating stats.
  - `src/pages/` : Contains all main views like `Dashboard` , `Library` , and `TodayRevisions` .
  - `src/components/` : Reusable UI elements like `Sidebar` , `Topbar` , and specialized `Charts` .
  - `index.css` : The global design system including color palettes and utility classes.
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## How to Run Locally

1. **Clone the repository** (if applicable)
2. **Install dependencies:**

```
npm install
```

3. **Start the development server:**

```
npm run dev
```

4. **Open your browser:** Navigate to the URL shown in your terminal (usually `http://localhost:5173` ).
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## Author

Developed by **Zeel Shah**. Helping developers build a smarter brain for DSA.